

$$g(x) = \frac{1}{\sqrt{x-|x|}}$$

Domain of $g(x) = x - |x| > 0$
 $\Rightarrow |x| > x$
 (Not possible as $|x|$ is always greater than or equal to x)

Solution for Q.no. 81 and 82:

Let
$$I = \int \frac{3 \cos x + 4 \sin x}{2 \cos x + 5 \sin x} dx$$

$$3 \cos x + 4 \sin x = A(2 \cos x + 5 \sin x) + B \left[\frac{d}{dx} (2 \cos x + 5 \sin x) \right]$$

$$3 \cos x + 4 \sin x = A(2 \cos x + 5 \sin x) + B(-2 \sin x + 5 \cos x)$$

Comparing coefficient of $\sin x$ and $\cos x$

$$3 = 2A + 5B \quad \dots(i)$$

$$4 = 5A - 2B \quad \dots(ii)$$

Solving (i) and (ii)

$$A = \frac{26}{29}, B = \frac{7}{29}$$

$$I = \int \frac{\frac{26}{29}(2 \cos x + 5 \sin x) + \frac{7}{29}(-2 \sin x + 5 \cos x)}{2 \cos x + 5 \sin x} dx$$

$$I = \int \frac{26}{29} dx + \frac{7}{29} \int \frac{(-2 \sin x + 5 \cos x)}{2 \cos x + 5 \sin x} dx$$

$$I = \frac{26}{29}x + \frac{7}{29} \ln(2 \cos x + 5 \sin x) + c$$

$$\alpha = 26, \beta = 7 \quad (\text{on comparing})$$

81. Option (d) is correct.

$$\alpha = 26$$

82. Option (a) is correct.

$$\beta = 7$$

Solution for Q.no. 83 and 84:

83. Option (d) is correct.

$$f(x) = \frac{x}{\ln(x)} \quad (x > 1)$$

$$f'(x) = \frac{\ln x - 1}{[\ln(x)]^2}$$

$$\begin{array}{c} \text{--- ve} \qquad \qquad \text{+ ve} \\ \hline 1 \qquad \qquad e \end{array}$$

$f'(x)$ is +ve, $x \in (e, \infty)$ i.e., $f(x)$ is increasing in interval $(e, \infty) \Rightarrow 1$ is correct.

$f'(x)$ is -ve, $\forall x \in (1, e)$ i.e., $f(x)$ is decreasing in the interval $(1, e) \Rightarrow 2$ is correct.

As $\ln x$ is an increasing function

$$7^9 > 9^7$$

$$\ln 7^9 > \ln 9^7$$

$$9 \ln 7 > 7 \ln 9$$

$\Rightarrow$ statement 3 is correct

$\therefore$ Statement 1, 2 and 3 are correct.

84. Option (d) is correct.

$$f'(x) = \frac{(\ln x)^2 \times \frac{1}{x} - (\ln x - 1) \times 2 \frac{\ln x}{x}}{(\ln x)^4}$$

$$= \frac{(\ln x)^2 - 2(\ln x)^2 + 2 \ln x}{x(\ln x)^4}$$

$$\Rightarrow f'(e) = \frac{1 - 2 + 2}{e \times 1} = \frac{1}{e} (> 0)$$

$\Rightarrow$ Statement 1 is correct

Since $f'(e) > 0$ (true)

$f(x)$ attains local minima at $x = e$.

$\Rightarrow$ Statement 2 is correct

A local minimum value occurs at $x = e$.

$$f(x) \text{ at } x = e, \text{ is } f(e) = \frac{e}{\ln e} = e$$

$\Rightarrow$ Statement 3 is correct

$\therefore$ Statement 1, 2 and 3 are correct.

Solution for Q.no. 85 and 86:

Given: $f(x)$ and $g(x)$ are two functions

$$g(x) = x - \frac{1}{x} \text{ and } fog(x) = x^3 - \frac{1}{x^3}$$

$$f\left(x - \frac{1}{x}\right) = x^3 - \frac{1}{x^3}$$

$$\Rightarrow f\left(x - \frac{1}{x}\right) = \left(x - \frac{1}{x}\right)^3 + 3\left(x - \frac{1}{x}\right)$$

85. Option (a) is correct.

$$f(x) = x^3 + 3x \quad \dots(i)$$

$$g[f(x) - 3x] = f(x) - 3x - \frac{1}{f(x) - 3x}$$

$$g[f(x) - 3x] = x^3 + 3x - 3x - \frac{1}{x^3 + 3x - 3x}$$

$$= x^3 - \frac{1}{x^3}$$

$$g[f(x) - 3x] = x^3 - \frac{1}{x^3}$$

86. Option (d) is correct.

$$f'(x) = 3x^2 + 3$$

$$f''(x) = 6x$$

Solution for Q.no. 87 and 88:

$$f(x) = |x| + 1 \text{ and } g(x) = [x] - 1,$$